

Supplementary Information: Validated Synthetic Patient Generation for Small Longitudinal Cohorts

Supplementary Figures

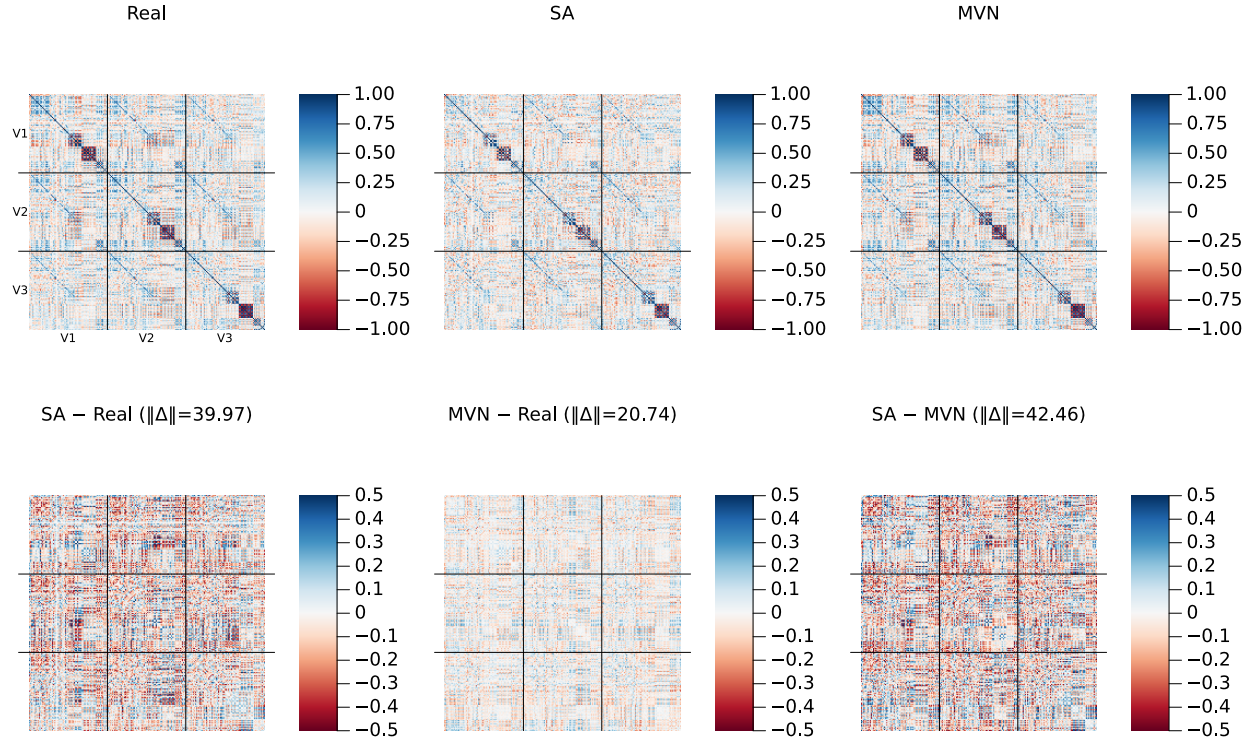


Figure S1: **Cross-visit correlation structure (amplified residual scale)**. Same layout as the main-text figure, but with the bottom-row residual panels plotted on a ± 0.5 color scale (vs. ± 1.0 in the main text) to reveal fine structure in the correlation differences. Frobenius norms of each residual matrix are shown in the panel titles. The MVN–Real residual is systematically smooth (reflecting regularization-induced shrinkage), while the SA–Real residual is noisier but unbiased.

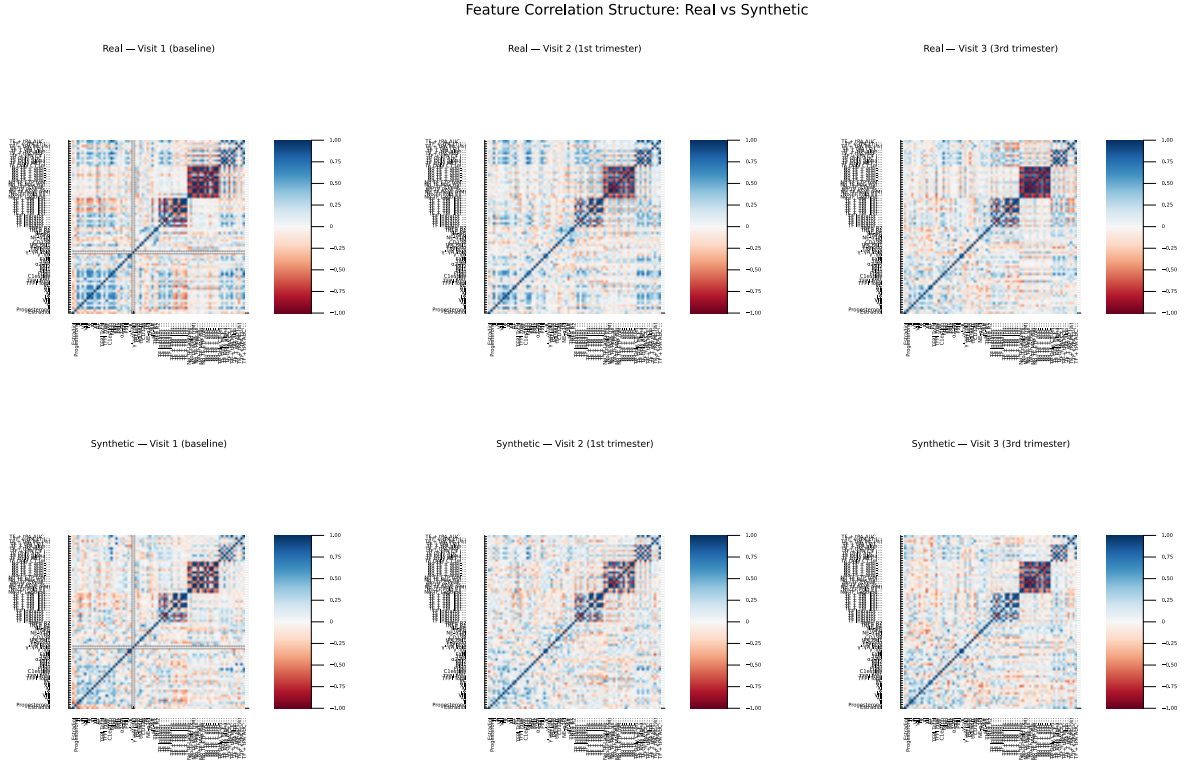


Figure S2: **Within-visit correlation heatmaps.** Pearson correlation matrices for real and SA-generated synthetic patients within each visit (72 features per visit). SA preserves the within-visit correlation structure across all three visits.

Supplementary Tables

Supplementary Figures

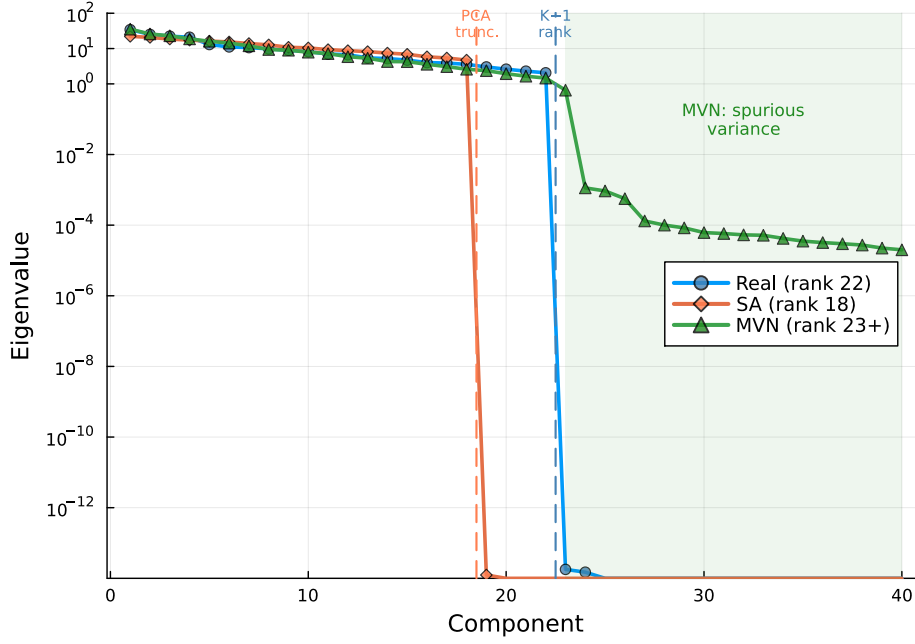


Figure S3: **Covariance eigenvalue spectrum.** Log-scale eigenvalues of the 216×216 sample covariance for Real ($K=23$, rank 22), SA ($N=100$, rank 18), and MVN ($N=100$) populations. Dashed vertical lines mark the SA PCA truncation point (component 18) and the data rank limit (component 22, $K-1$). SA’s spectrum drops cleanly at the PCA truncation, matching the low-rank data structure. MVN’s Ledoit–Wolf regularization inflates eigenvalues beyond component 22 (green shaded region), introducing spurious variance in 194 dimensions where the data contain none. This spurious variance is the structural origin of the increased MVN dispersion visible in the PCA projections (main-text Fig. 4).

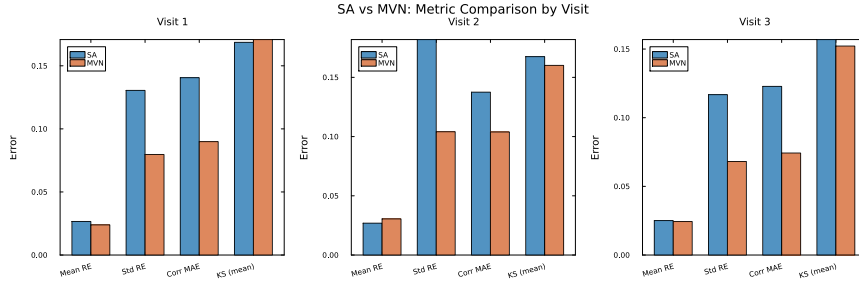


Figure S4: **SA vs. MVN quantitative comparison.** Per-visit comparison of mean relative error, standard deviation relative error, KS statistic, and correlation MAE between SA and MVN generation methods. Both methods perform comparably on marginal statistics (Level 1), but diverge on structural measures (correlation MAE).

Table S1: **Inverse temperature sensitivity analysis.** SA generation quality as a function of β/β^* , where $\beta^* \approx 5.27$ was identified via entropy inflection in the 18-dimensional PCA space ($K=23$ stored patterns). The theoretical prediction $\beta^* = \sqrt{d} = \sqrt{18} \approx 4.24$ is close to the empirical value. Operating at $\beta = \beta^*$ (bold row) balances generation novelty against fidelity: lower β produces more diverse but less faithful samples; higher β produces more faithful but less novel samples (approaching memorization). MRE = median relative error of per-feature means; Corr MAE = mean absolute error of the correlation matrix; Novelty = $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$.

β/β^*	β	Noise scale	PC1 std ratio	Novelty	MRE (%)	Corr MAE
0.1	0.53	0.436	0.455	0.501	3.6	0.242
0.2	1.05	0.308	0.343	0.495	2.8	0.236
0.3	1.58	0.252	0.297	0.488	2.4	0.230
0.5	2.63	0.195	0.252	0.476	2.1	0.220
0.7	3.69	0.165	0.230	0.463	1.9	0.212
1.0	5.27	0.138	0.212	0.444	1.8	0.200
1.5	7.90	0.113	0.197	0.415	1.5	0.183
2.0	10.53	0.097	0.184	0.389	1.6	0.173
3.0	15.80	0.080	0.174	0.336	1.4	0.154

Table S2: **Multiplicity weighting parameters for conditional generation.** For each clinical subgroup, the multiplicity ratio ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated patterns. K_{eff} is the participation ratio, quantifying the effective number of patterns contributing to generation. PCOS required the highest ρ (26.7) due to having only 3 stored patterns, reducing K_{eff} to 4.6 and explaining the larger MREs observed for this subgroup at Level 3.

Condition	K_{sub}	K_{bg}	ρ	f_{eff}	K_{eff}
Healthy	14	9	2.57	0.80	19.9
PCOS	3	20	26.67	0.80	4.6
PE	5	18	14.40	0.80	7.7

Table S3: **SA generation hyperparameters.** All parameters used for synthetic patient generation. The same values were used for unconditioned and conditioned generation except where noted.

Parameter	Value	Description
α	0.05	Langevin step size
T	2,000	Langevin iterations per sample
β^*	5.27	Inverse temperature (entropy inflection)
PCA threshold	95%	Variance retained
d_{PCA}	18	PCA dimensionality
N	100	Synthetic patients per cohort
Random seed	42	For reproducibility
f_{target}	0.80	Target attention fraction (conditioned only)

Table S4: **BZ2012 ODE model calibration.** Five rate constants were calibrated on Visit 1 real patients ($n=23$) using bounded Nelder–Mead optimization (Fminbox) in log-scale-factor space. The cost function was the mean normalized relative squared error across five TGA features (lagtime, peak, time-to-peak, maximum rate, ETP) under both TF-only and TF+TM conditions. Bounds: $\pm \log(100)$ ($0.01\times$ to $100\times$ default). Convergence: $f_{\text{tol}} = 10^{-6}$, 500 iterations.

Rate constant	Role	Default	Calibrated	Scale
prothrombinase k_{cat}	Thrombin burst	63.5 s^{-1}	21.94 s^{-1}	$0.35\times$
intrinsic Xase k_{cat}	Amplification Xa	8.2 s^{-1}	0.169 s^{-1}	$0.021\times$
extrinsic Xase k_{cat}	Initiation Xa	6.0 s^{-1}	93.2 s^{-1}	$15.5\times$
PC activation k_{cat}	Protein C activation	0.41 s^{-1}	0.890 s^{-1}	$2.17\times$
mIIa conversion k	mIIa $\rightarrow$ IIa	2.3×10^8	1.15×10^9	$5.01\times$

Units for mIIa conversion: $\text{M}^{-1}\text{s}^{-1}$

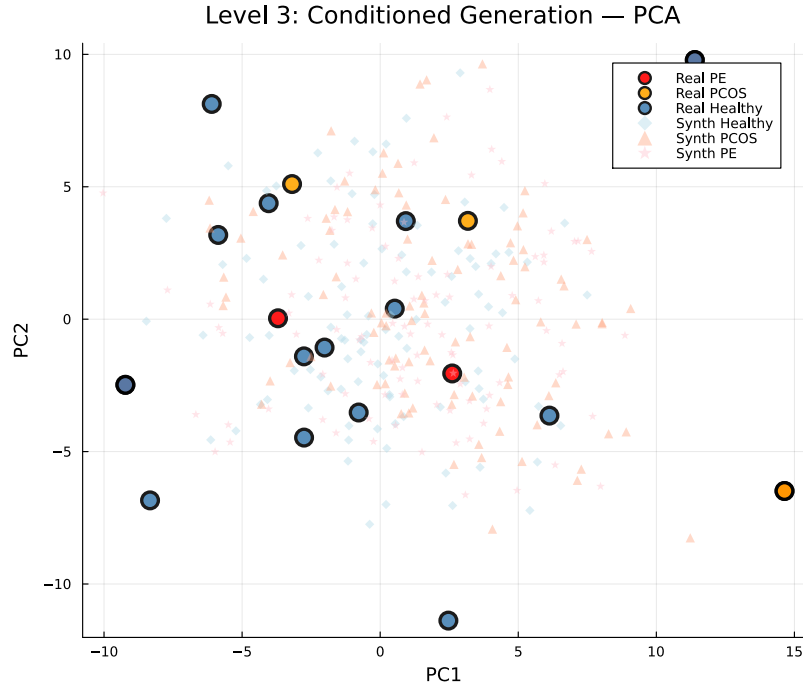


Figure S5: **Conditioned generation: PCA projection (pooled).** PCA projection of condition-specific synthetic cohorts generated by multiplicity-weighted SA. Large markers: real patients (blue = Healthy, $n=14$; orange = PCOS, $n=3$; red = PE, $n=5$). Small markers: synthetic patients ($N=100$ per condition). With only 3 PCOS and 5 PE patients, the subgroups do not form clearly separable clusters in the first two principal components, but the synthetic cohorts concentrate around their respective real counterparts.

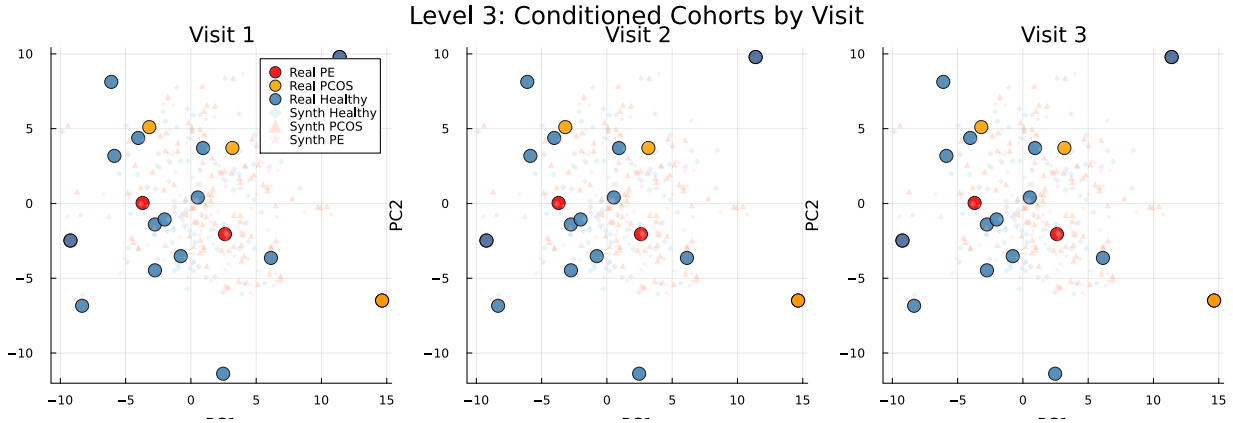


Figure S6: **Conditioned generation: PCA projection by visit.** Same as Supplementary Fig. S5 but separated by visit (V1, V2, V3). The condition-specific clustering patterns are maintained within each individual visit.

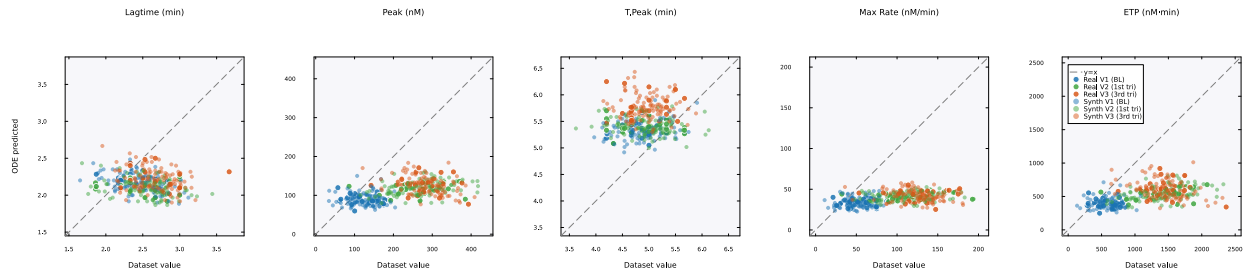


Figure S7: **Level 4: Mechanistic model predicted vs. measured TGA features (TF+TM condition).** Same layout as main-text Fig. 6. The model systematically underpredicts peak and maximum rate ($\approx 0.5 \times$ measured) for both real and synthetic patients, reflecting overcorrection of the protein C anticoagulant pathway. The shared bias indicates that synthetic patients are processed identically to real patients by the mechanistic model.

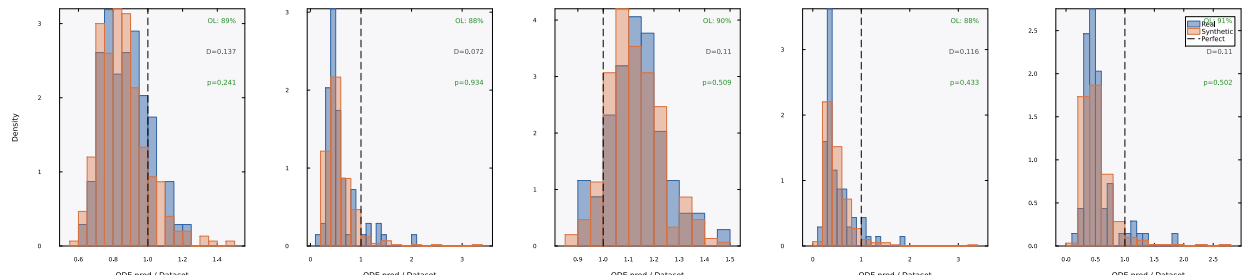


Figure S8: **Level 4: Predicted/measured ratio distributions (TF+TM condition).** Same layout as main-text Fig. 7. Despite the systematic bias in peak and maximum rate, the ratio distributions for real and synthetic patients overlap substantially (88–91% cloud overlap).

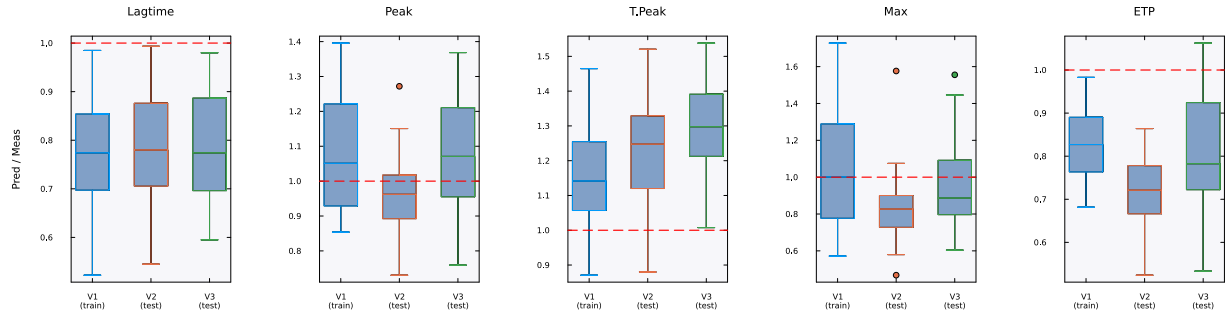


Figure S9: **Level 4: Cross-visit calibration generalization (TF-only).** Boxplots of the predicted/measured ratio by visit for real patients. The model was calibrated on Visit 1 (“train”) only. Ratios near 1.0 at Visits 2 and 3 indicate that the population-level calibration generalizes across pregnancy timepoints.

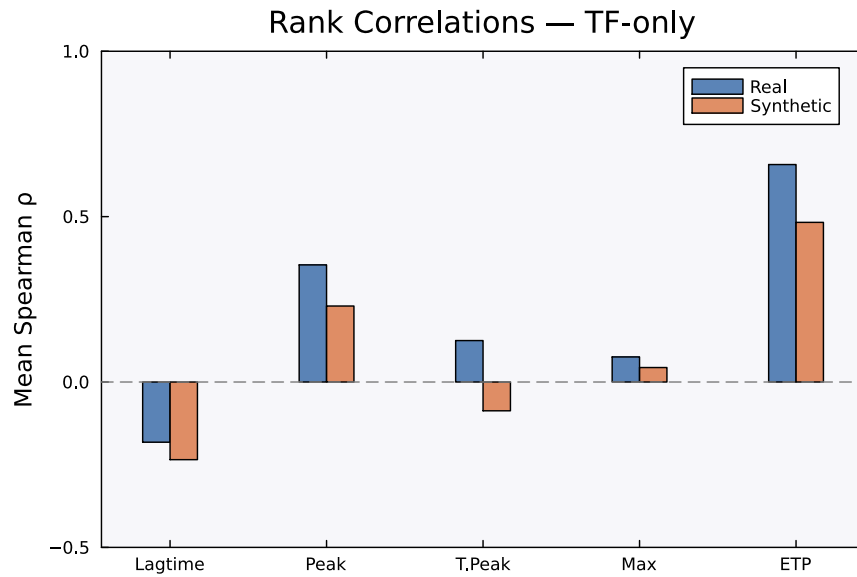


Figure S10: **Level 4: Spearman rank correlations (TF-only).** Mean Spearman ρ between predicted and measured TGA features, averaged across visits, for real and synthetic patients. ETP is the best-predicted feature ($\rho \approx 0.6$ – 0.8). Weak rank correlations for other features are expected given the population-level (not patient-level) calibration.

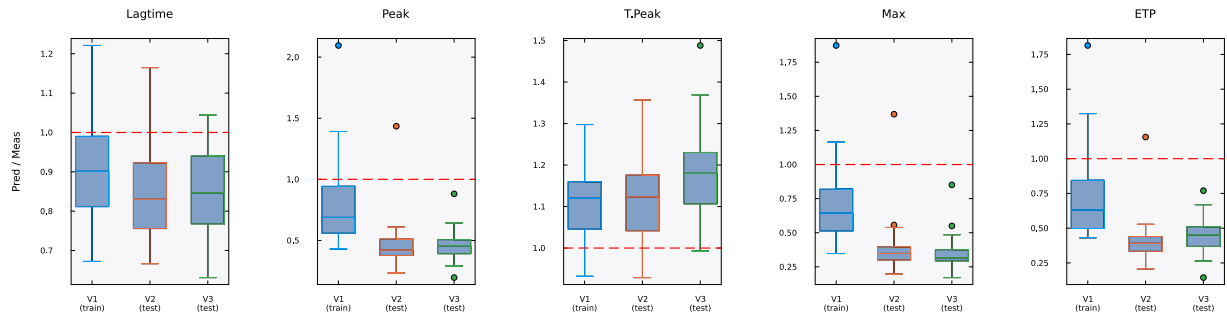


Figure S11: **Level 4: Cross-visit calibration generalization (TF+TM).** Same layout as Supplementary Fig. S9. The TF+TM calibration does not generalize as well as TF-only, particularly for peak and maximum rate, due to overcorrection of the protein C pathway.

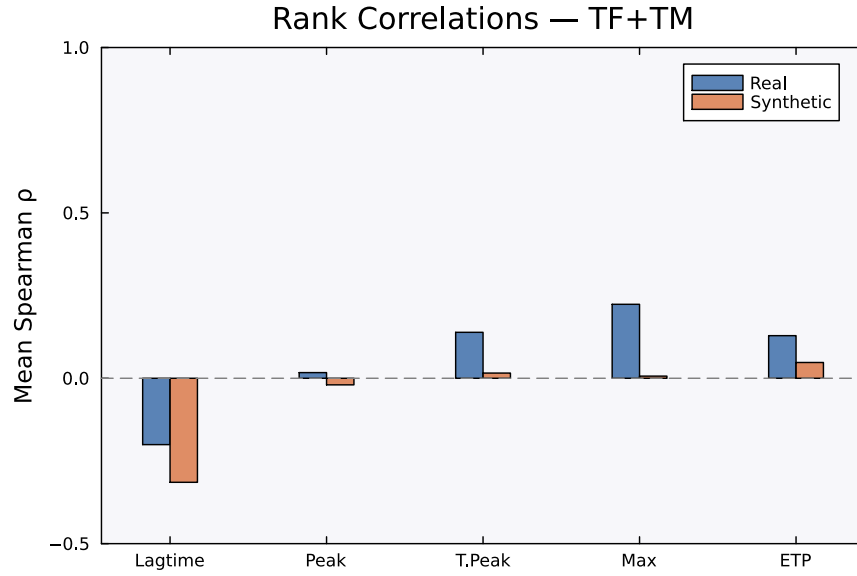


Figure S12: **Level 4: Spearman rank correlations (TF+TM)**. Same layout as Supplementary Fig. S10. Rank correlations under TF+TM are generally weaker than TF-only, consistent with the poorer calibration generalization under this condition.

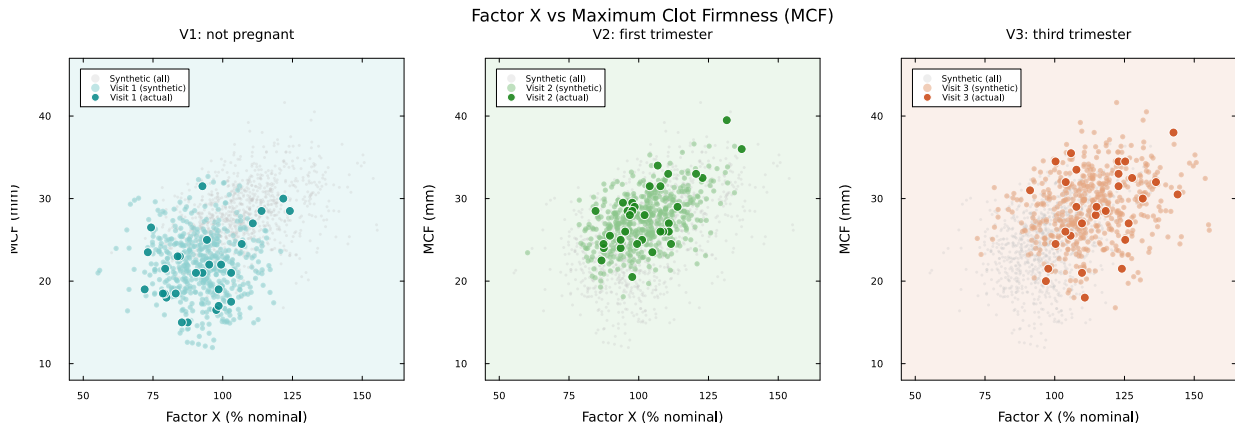


Figure S13: **Factor X vs. Maximum Clot Firmness (MCF) trajectory**. The characteristic pregnancy-driven shift in the Factor X–MCF plane is preserved in SA-generated synthetic patients, demonstrating that biologically meaningful multivariate trajectories are captured by the generation pipeline.

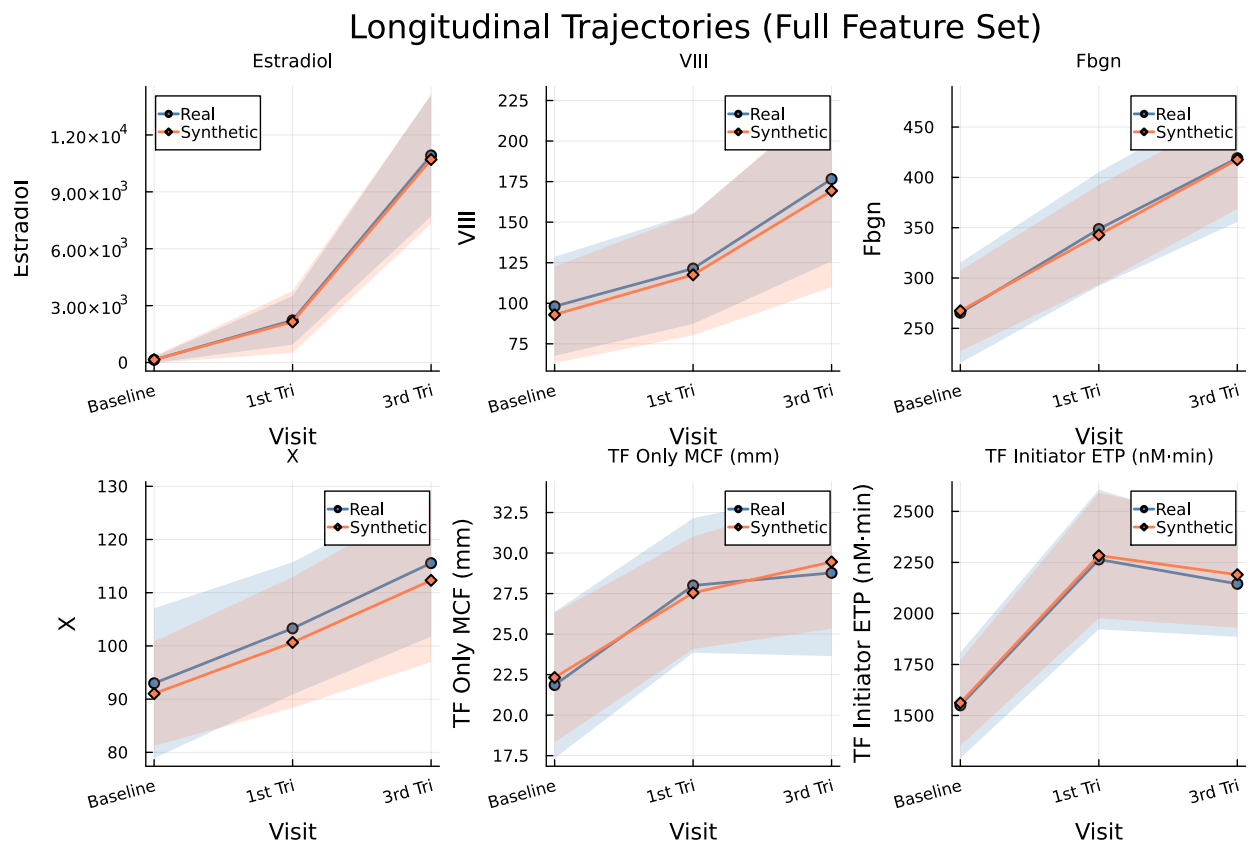


Figure S14: **Full longitudinal trajectories.** Individual patient trajectories (all 72 features) across three visits for both real and SA-generated synthetic patients, demonstrating coherent longitudinal patterns.